Checking your Docker system

Get information about Docker

- `rpm -q` *`pkg`* `or dpkg-query --show` *`pkg`*
- `systemctl status` *or* `service docker.io status`
- `docker info` — *System-wide Docker information*
- `docker version` — *Docker version information*
- `docker top` — *Show info on docker processes*
- `docker diff` — *Show container filesystem changes*
- `docker history` — *Show container build history*

Removing Containers and Images

Containers and images continuously consume storage. To prevent that:

- Prevent containers during build (`docker build -rm`)
- Don't save a container after a run (`docker run -rm`)
- Remove containers you no longer need (`docker rm`)
- Remove images you no longer need (`docker rmi`)
- Run a script to remove all containers or images at once

Choose what Goes into a Dockerfile File

These instructions can go in your Dockerfile file:

- **FROM**: Identifies the base image (must be first instruction)
- **MAINTAINER**: Identifies author in Author field for the image
- **RUN**: Executes a command while building an image
- **CMD**: Sets command to execute when container starts
- **EXPOSE**: Exposes container ports to host at runtime
- **ENV**: Sets environment variable to pass to runtime command
- **ADD**: Copies files into the container.

Choose what Goes into a Dockerfile File (cont.)

- **`ENTRYPOINT`**: Sets container to run as chosen executable
- **`VOLUME`**: Mounts storage from host into the container
- **`USER`**: Identifies the user assigned to running the container
- **`WORKDIR`**: Sets current working directory for `RUN`, `CMD` and `ENTRYPOINT` commands run in the container
- **`ONBUILD`**: Used when building base images, to add instructions a subsequent build would use to add code or data to the image that were not available during the initial base build

General Rules for Building Containers

- Use root privilege to build containers, non-root user to run
- Choose standard base images
- Pull specific tag when pulling an image to:
  - Use less space
  - Identify exact image used
- Use ENV instructions to store useful information
- Restrict container to one process (no extra access like ssh)
- Don't use / as your build directory

Using Software Repositories

- Use tested, well-know repositories
- Use software packaged in standard formats (deb, rpm, etc.)
- Clean up install cache to keep containers lean
- If available, use keys to check packages you install
- To add repositories to install within a container:
  - **Fedora**: add .repo file /etc/yum.repos.d
  - **Ubuntu**: add .list file to /etc/apt/sources.list.d/
  - **RHEL**: Containers get subscriptions from build host

Accessing Networking from Containers

- Docker starts a private network (such as 172.17.42.0/16)
- The docker0 network routes information:
  - Between containers
  - To other network interfaces on host (set to packet forward)
- Containers are assigned IP addresses on that network
- Containers in the same pod can connect via local host
- Docker networking initiatives are working to make network configurations that span multiple hosts

Base Images

- Small copy of an operating system in a container
- Basic tools and software tools (yum, apt-get)
- Provided by Linux distros (standard and bare bones)
- Examples include official container repositories:
  - **Ubuntu**: https://hub.docker.com/_/ubuntu/
  - **Fedora**: https://hub.docker.com/_/fedora/
  - **CentOS:** https://hub.docker.com/_/centos/
  - **Red Hat Enterprise Linux**: https://access.redhat.com/containers#/category/Base%20Image

Builder Images

- Include operating system plus runtime components
- Developers can add their own source code
- Builder images for different versions of:
  - Python
  - Node.js
  - JBoss Realtime Decision Server, Intelligent Process Server, Web Server (Tomcat),
  - Ruby
  - Apache
- Source-to-image (s2i) injects source code in containers

Custom Base Images

- Allows complete control of image contents
- Has no parent container image
- Created from a software tarball or by just adding executable
- Either FROM scratch or no FROM line
- Instructions for creating custom base images:

https://docs.docker.com/engine/userguide/eng-image/baseimages/

- Example: Creating a CentOS base image:

https://github.com/moby/moby/blob/master/contrib/mkimage-yum.sh